

PCA test

May 13, 2026

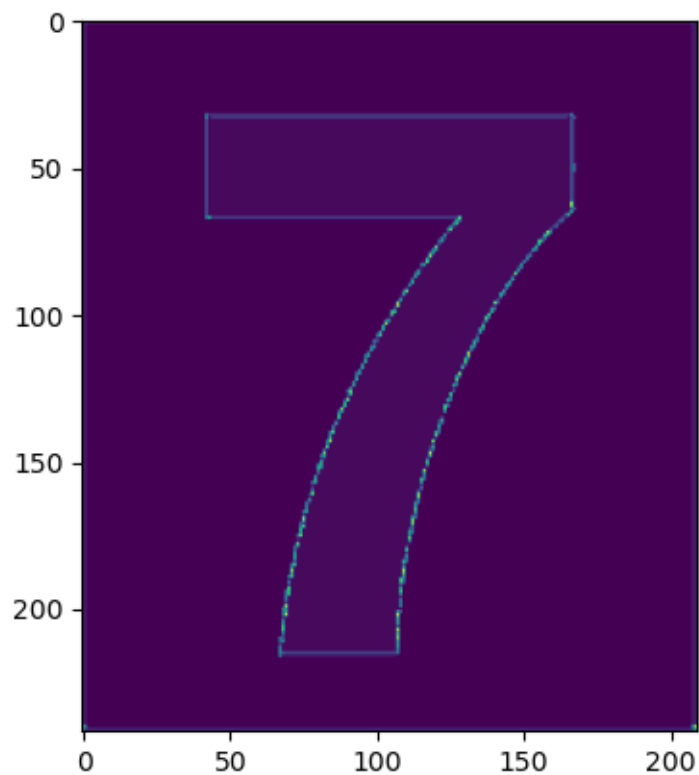
```
[25]: from PIL import Image
import numpy as np
import matplotlib.pyplot as plt

# Load the image
img = Image.open('images.png')

# Convert to NumPy array
img_array = np.asarray(img)
```

```
[26]: plt.imshow(img_array)
```

```
[26]: <matplotlib.image.AxesImage at 0x1d62a50a4e0>
```



```
[30]: import pca
```

```
[32]: model = pca.PCA()
```

```
[33]: X=img_array
```

```
[34]: X.shape
```

```
[34]: (242, 209)
```

```
[35]: X.shape
```

```
[35]: (242, 209)
```

```
[36]: X
```

```
[36]: array([[ 1,  3,  3, ...,  3,  3,  1],
          [ 3,  5,  2, ...,  2,  5,  3],
          [ 3,  2,  0, ...,  0,  2,  3],
          ...,
          [ 3,  2,  0, ...,  0,  2,  3],
          [36, 12,  0, ...,  0, 12, 36],
          [20,  7,  4, ...,  4,  7, 20]], dtype=uint8)
```

```
[37]: model.fit_data(X)
```

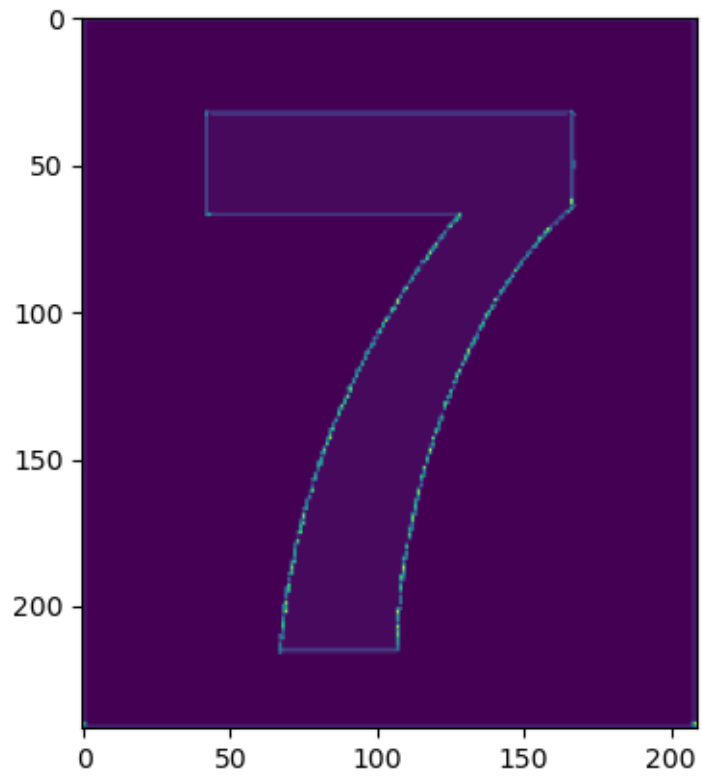
Dataset fitted successfully

```
[38]: ldd = model.compress(100)
```

The data projected into the low dimensional PCA space of 107 dimension

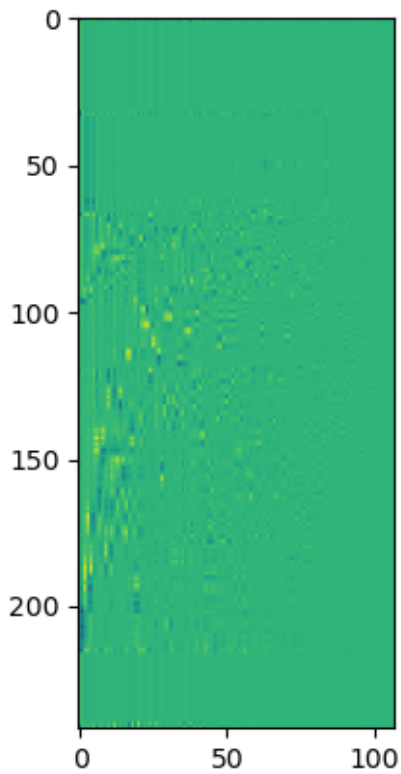
```
[43]: rci = model.represent()
      plt.imshow(rci)
```

```
[43]: <matplotlib.image.AxesImage at 0x1d62bbd6990>
```



```
[44]: plt.imshow(com)
```

```
[44]: <matplotlib.image.AxesImage at 0x1d62bba3740>
```



```
[47]: error = np.sum((X - rci)**2)
      error
```

```
[47]: 5.487549249310488e-24
```

```
[49]: print(np.allclose(X,rci))
```

```
True
```

```
[ ]:
```

```
[ ]: tt =np.array( [[2,4,3],[5,0,1]])
      np.cov(tt)
```

```
[ ]: rng = np.random.default_rng()
      arr_2d = rng.random((124, 887))
```

```
[ ]: b = p.PCA()
      b.fit_data(arr_2d)
```

```
[ ]: q = b.represent(100)
```

```
[ ]: error = np.sum((arr_2d - q)**2)
      error
```

```
[ ]: print(np.allclose(arr_2d, q))
```

```
[ ]:
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[ ]:
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